

MARS TELECOMMUNICATIONS NETWORK (MTN)
QUALITY ASSURANCE SURVEILLANCE PLAN
(QASP)

1.0 INTRODUCTION

1.1 Purpose

The purpose of this Government Quality Assurance Surveillance Plan (QASP) is to define the overall framework the NASA Goddard Space Flight Center (GSFC) intends to use to monitor and survey Contractor performance under the Mars Telecommunications Network (MTN) Contract No. **TBD**. This Firm-Fixed-Price (FFP) contract provides for the Government to obtain a Mars telecommunications system, including all associated development, integration, test, launch, and operational services necessary to achieve full operational capability at Mars and provide relay services for five Earth-years.

The QASP is a Government-developed surveillance tool prepared in accordance with FAR 46.601 and NFS 1846.401. It is not part of the contract, per NFS 1846.401(a)(iii), but is provided to the Contractor for informational purposes only.

1.2 Scope

This QASP has been developed to evaluate Contractor performance while executing the Performance Work Statement (PWS), which is delivered by the Contractor and incorporated into the contract upon award. It is designed to provide an effective surveillance framework for monitoring Contractor performance, deliverables, processes, and products provided under this MTN contract across all three contract phases:

- (1) Development through Implementation**
- (2) Launch through Relay Commissioning at Mars**
- (3) Relay Operations**

This QASP defines the process the Government will follow to obtain data, evaluate the Contractor, and determine if contract performance conforms to contract requirements. The Contractor, and not the Government, is responsible for management and quality control actions to meet the terms of the contract.

The role of the Government is quality assurance to ensure contract standards are achieved. The Contractor will provide Government insight into its program of inspections and monitoring actions. The first major step toward ensuring a self-correcting contract is to confirm that the quality control program defined at the beginning of the contract provides

sufficient measures to establish high confidence of success. Once the quality control program is understood, careful application of the processes and standards will ensure a robust quality assurance program.

This plan is applicable to all work performed and all products and deliverables provided by the MTN Contractor. Throughout this QASP, the term "Contractor" is used. Unless explicitly stated otherwise, the term "Contractor" applies to both the MTN prime Contractor and any and all of its subcontractors.

2.0 FORMS OF SURVEILLANCE

2.1 Surveillance Strategies

Insight

Insight is an assurance process that uses performance requirements and, where definable, performance metrics to ensure process capability, product quality, and end-item effectiveness. Insight relies on gathering a set of product or process data that provides adequate visibility into the integrity of the product or process. Data may be acquired from Contractor records, typically in a non-intrusive parallel method.

Insight as applied to this contract will allow the Contractor to assume increased responsibility and accountability for the integrity of processes. Insight will rely heavily on evaluating planned contract deliverables, performance standards, and existing Contractor procedures and working documents. The specific mechanisms and processes by which the Government will exercise insight are defined in the Contractor-delivered DRD-PM-08, Insight and Oversight Implementation Plan.

Oversight

The Government will gather information pertaining to the Contractor's processes through direct involvement and/or inspection and will monitor those processes directly. The Government's exercise of oversight will be determined necessary by the Contracting Officer's Representative (COR).

The Government may use a variety of surveillance methods to evaluate Contractor performance. Government surveillance will be conducted on a non-interference basis and in a manner that will not substantially delay work being performed by the Contractor. The specific oversight mechanisms and processes are further defined in the Contractor-delivered DRD-PM-08, Insight and Oversight Implementation Plan. Government oversight will be applied to a limited set of activities and processes with high impact to mission success such as fault tolerance, subsystem and system level testing and acceptance, deviations from plans, critical event planning, risk management, and problem/anomaly diagnosis and resolution.

2.2 Surveillance Team

All surveillance activities will be implemented using NASA and NASA support-contractor personnel, a delegated agency (e.g., Defense Contract Management Agency (DCMA)), and/or surveillance support personnel. The surveillance team will be composed of:

- MTN Project Manager
- MTN Engineering Personnel
- MTN Procurement Personnel (i.e., CO, Contract Specialist)
- MTN Program Personnel (i.e., COR)
- MTN Safety and Mission Assurance Personnel
- MTN Information Technology Security Personnel
- MTN Property Administrator Personnel
- MTN Project Planning and Control Lead

The MTN Project Manager is responsible for the overall programmatic and technical health of the MTN contract. The MTN Project Manager serves as the senior Government representative for day-to-day program execution, provides leadership and direction to the surveillance team, and maintains visibility into Contractor performance across all contract phases. The MTN Project Manager, in collaboration with the CO and COR, is responsible for preparing and submitting regular Contractor Performance Assessment Reporting System (CPARS) evaluations, which will be reviewed by the Contractor and will become part of the Past Performance Information Retrieval System (PPIRS).

MTN Engineering Personnel, in coordination with the Project Manager, are responsible for exercising insight and oversight of the technical aspects of the contract. Engineering personnel perform these functions through regular interaction with the Contractor in technical meetings, particularly those related to key decisions, risks, and evaluation of milestone deliveries.

The MTN CO is responsible for ensuring performance of all necessary actions for effective contracting, ensuring compliance with the terms of the contract, and safeguarding the interests of the United States in its contractual relationships. The CO will also assure that the Contractor receives impartial, fair, and equitable treatment under this contract. The CO is ultimately responsible for the final determination of the adequacy of the Contractor's performance.

The MTN COR is designated in writing by the CO to act as his/her authorized technical representative to assist in administering the contract. The COR monitors the technical work performed under the contract, evaluates Contractor performance, serves as the primary interface between the Contractor and the CO for all technical matters, reports on contract status to Project Management, and recommends corrective action when

necessary. The COR is not empowered to make any contractual commitments, authorize any contractual changes on the Government's behalf, or in any way direct the Contractor to operate in conflict with the contract terms and conditions. Any changes that the Contractor deems may affect the contract price, terms, or conditions shall be referred to the CO for action. The COR's limitations of authority are contained in the NASA Form 1634, COR Delegation.

MTN Safety and Mission Assurance Personnel will monitor the Contractor's compliance with applicable safety and health requirements throughout all phases of the MTN program, including development, integration, test, launch, and operational activities. Surveillance will be conducted on a non-interference basis consistent with the overall surveillance strategy defined in this plan.

MTN Information Technology Security Personnel will monitor the Contractor's compliance with applicable IT security requirements throughout the MTN program lifecycle. IT security surveillance will be conducted in alignment with the IT systems security framework and applicable NASA and NIST cybersecurity standards.

MTN Property Administrator will monitor and administer Government property, if any, provided to or acquired by the Contractor under this contract, in accordance with applicable FAR and NASA FAR Supplement property administration requirements.

MTN Project Planning and Control Lead applies financial oversight in the context of a firm, fixed price (FFP) contract inclusive of evaluation of milestone deliveries, implementation of change orders, and other schedule or price related matters.

2.3 Communications

The Government and Contractor will maintain regular communication throughout all phases of the MTN contract. Communication methods will include, but are not limited to:

- Periodic technical and programmatic reviews
- Technical Interchange Meetings (TIMs)
- Electronic mail
- Informal discussions
- Other communication methods as required

The specific engagement mechanisms, touchpoints, and communication structures are further defined in the Contractor-delivered DRD-PM-08, Insight and Oversight Implementation Plan, and are aligned with the NASA Engagement Plan.

2.4 Evaluation and Reporting

Government evaluation of Contractor performance will be milestone and deliverable-based, consistent with the FFP contract structure. Evaluation and reporting activities will include:

- Review and acceptance of contract deliverables at defined milestones
- Assessment of Contractor performance against the milestone payment schedule
- Evaluation of technical progress at key program reviews and decision gates
- Inspection of deliverable hardware, software, and documentation
- Review of Contractor-submitted reports and data products

2.5 Work Review and Performance Monitoring

The Government will evaluate the Contractor's accomplishment of technical and programmatic objectives through the following activities:

- Periodic meetings with the Contractor to review progress and identify and resolve problems and risks
- Review of Contractor milestone completion and deliverable acceptance across all three contract phases
- Participation in key technical reviews, including design reviews, test and qualification reviews, and operational readiness reviews
- Access to facilities and testing activities as appropriate, consistent with DRD-PM-08
- Participation in risk management meetings
- Participation in problem, failure, and anomaly review boards
- Preparation and submission of Contractor Performance Assessment Reporting System (CPARS) reports
- Conduct of inspection and acceptance of Contractor deliverables and hardware
- Review of Contractor safety procedures and performance
- Validation that the Contractor is performing in accordance with required contractual plans (e.g., testing, integration, quality assurance) through physical inspection, review of submitted reports, and meetings with the Contractor's program manager

2.6 Record Keeping and Documentation

The COR is responsible for maintaining accurate and complete records of the Contractor's performance. The COR is responsible for collecting performance evaluation documentation and maintaining any other material that reflects the quality of Contractor performance. Even though the Contractor is responsible for ensuring performance compliance under their quality control requirements, the Government will give the Contractor prompt notice of deficiencies observed by the COR. This does not imply that the Government is obligated to carry out quality control inspections for the Contractor, but

that on discovering a problem, the Government will inform the Contractor in detail of the incident. The COR will keep the CO informed of any issues brought to the attention of the Contractor.

If performance issues arise, they will be discussed immediately between the CO, COR, and Contractor. Action plans will be developed and resolutions tracked for closure by the Contractor and Government.

During the course of the contract, the COR shall retain a copy of all surveillance documentation. At the end of each surveillance period, these records shall be forwarded to the Contracting Officer for inclusion in the contract file. When a specific deliverable or milestone becomes unsatisfactory during a surveillance period, a copy of the documentation supporting the unsatisfactory performance shall be forwarded to the Contracting Officer for action between the Government (CO and COR) and the Contractor.

2.7 Contract Administration

The Contractor will maintain effective communication within its own organization, including subcontractors. Problems, technical issues, and programmatic changes will be promptly communicated to all concerned parties. Prime Contractor communication with subcontractors will be seamless and transparent to the Government.

3.0 CRITERIA FOR EVALUATING PERFORMANCE

The Contractor's performance across all areas of the MTN contract will be monitored to assure that ineffective or wasteful methods are not being used. As required by FAR 42.1502 and GPR 5100.2, Supplier Performance Evaluations, the CO, in collaboration with the COR, will complete Contractor Performance Assessment Reporting System (CPARS) evaluations on a regular basis, aligned with MTN program phases and milestones. CPARS evaluations will also be reviewed by the Contractor and will become part of the Past Performance Information Retrieval System (PPIRS).

Performance will be evaluated against the following criteria:

- **Schedule Performance:** The Contractor's ability to meet milestone dates, deliver hardware and software on schedule, and maintain the required hardware delivery date of December 31, 2028, and full operational readiness at Mars by end of calendar year 2030
- **Technical Performance:** The Contractor's ability to meet performance requirements specified in Attachment J-01 across all four MTN mission objectives
- **Cost Control:** The Contractor's ability to execute within the FFP contract value and manage program risks that could affect cost
- **Management Effectiveness:** The Contractor's ability to manage the program, subcontractors, risks, and issues in a transparent and effective manner

- **Quality Assurance:** The Contractor's adherence to the quality control program, standards, and verification plans
- **NASA Engagement:** The Contractor's effectiveness in providing NASA with the insight and visibility necessary to support informed decision-making, consistent with DRD-PM-08

In the event of a finding that requires immediate attention, the COR will notify the CO of the situation for necessary action. The Contractor will be required to submit a written response and any corrective action steps needed.

4.0 SUMMARY

This Government Quality Assurance Surveillance Plan defines the framework GSFC will use to monitor the MTN contract and assure that the Contractor performs in accordance with the terms and conditions of the contract. GSFC will employ an insight-based surveillance strategy, as further defined in the Contractor-delivered DRD-PM-08, Insight and Oversight Implementation Plan. The goal is to balance the level of Government surveillance with the perceived impacts and risks of mission failure across all three phases of the MTN program, while enabling the Contractor the flexibility to execute the program efficiently and meet the aggressive schedule requirements.